On locality conditions for allomorphy and the ordering of interface operations*

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1. Introduction

Contextual allomorphy (henceforth, allomorphy) is a phenomenon whereby the form of a morpheme is determined based on some property of another morpheme in its environment. This paper discusses *suppletive* allomorphy, where the different surface forms of a morpheme are not plausibly derived from a single underlying form by (morpho)phonological rules. A typical example of this kind of allomorphy in English are the various different exponents of the plural morpheme, including *-s* (*cats*), *-en* (*oxen*), and -∅ (*sheep*).

Research in Distributed Morphology (DM) has found that many patterns of allomorphy can be explained if allomorph selection is done cyclically (see Marantz 2001, 2007, a.o; Bešlin 2025 for a recent overview). Categorizers (*v*, *n*, *a*) are the relevant cyclic heads, and cyclic spell-out is governed by the principle in (1), reminiscent of (but not identical to) the weak Phase Impenetrability Condition (Chomsky 2001).

(1) Schematization of cyclic domains (Embick 2014:272):

a. Cyclic *y* merged in [*y* [X [Y [*x* $\sqrt{\text{ROOT}}$...]]]]

b. Cyclic domain centered on *x* = [X [Y [*x* $\sqrt{\text{ROOT}}$]]] sent to interfaces

Per (1), for purposes of allomorph selection, the root can interact with the first cyclic head *x* and any non-cyclic heads in *x*'s extended projection (X, Y). The root cannot interact with the second cyclic head *y* because they are in separate spell-out domains.

Several ancillary mechanisms have been argued to further delimit allomorphy; they include (i) linear adjacency (Embick 2010, a.o.); (ii) structural adjacency (Adger, Béjar, and Harbour 2001, 2003, a.o.); and (iii) accessibility domains, a claim that roots are visible up to the first node above the first categorizer–Y, but not X in (1) (Moskal 2015).

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In this paper, I show that there are violations of (i)-(iii) in Bosnian/Croatian/Serbian (BCS) allomorphy. In section 2, I show that *n*(ominalizer)s in BCS complex nominals show allomorphic conditioning by a linearly non-adjacent morpheme. In section 3, I show that the root and comparative affix are visible to each other in BCS negative comparative adjectives despite the structural intervention of the *a*(djectivizer) and negation. Finally, in section 4 I discuss the implications of the evidence reported here and in the existing literature for the timing of Vocabulary Insertion (VI) and linearization.

2. Case study I: Complex nominals

In this section, I show that in some BCS complex nominals the choice of *n*-allomorph is sensitive to the identity of the root in the string ROOT-AV-TH-*n*, in clear violation of the linear adjacency requirement. Consider the BCS morpheme *-av* in (2), used productively to produce imperfective (atelic) forms of perfective (telic) verbal stems.

(2) a. prod-a-ti
sell-TH-INF
'sell'

b. prod-**av**-a-ti
sell-AV-TH-INF
'be selling'

There is substantial controversy in the Slavic literature over (i) the segmentation of the morphemes AV and TH, and (ii) the identity of the morpheme *-av*. While many sources treat AV and TH in examples like (2b) as one morpheme (e.g., Tatevosov 2015, Biskup 2022), others segment them as I have done here (e.g., Łazorczyk 2010, Quaglia et al. 2022), preserving the presence of the theme vowel in both verb forms in (2). What is relevant for my purposes here is that *-av* (with or without the following theme vowel) is a morpheme, that is, a syntactic terminal.[1] The second controversy is about the identity of *-av*. While that morpheme has traditionally been called a *secondary imperfectivizer* given its role in the verbal domain, there are more recent proposals that it is an *eventizer* (Tatevosov 2015), or a root (Quaglia et al. 2022). I will keep glossing it simply as AV, but see section 2.1 where I show that the proposal to treat *-av* as a root receives support from stress patterns.

We can now turn our attention to agent nominals in BCS. BCS agent nominals come in varying degrees of morphological complexity; compare the forms in (3) with those in (4)-(6), where (4)-(6) contain the morpheme AV and (3) does not. As these examples show, BCS agent nominals can have a number of different *n*-allomorphs, including *-telj* (often interchangeable with *-lac*), *-ač*, and *-ac*.[2] The *n*-allomorphs in (3)-(6) are not interchangeable (e.g., **gledac*, **poznavač*, **predsedavac*, **prodavatelj*, **proučavač*, **predavac*).

[1]Given the controversy regarding the segmentation of AV and TH, I will (conservatively) not treat the theme vowel as an intervener in the discussion that follows. If the theme vowel were a separate morpheme, this would only strengthen the conclusions made here. Some theme vowels have been argued to be dissociated morphemes (Oltra-Massuet 1999), but dissociated morphemes are inserted prior to VI (Embick 1997), so they should still count as potential interveners for allomorphy.

[2]The theme vowel cannot be detected in the surface form of *-ač* and *-ac* nouns, because the suffix begins with a vowel. In Slavic, this leads to deletion of the first (leftmost) vowel, here the theme vowel (Jakobson 1948). This deletion is a phonological process and it should therefore happen after VI. However, it is not crucial that the theme vowel still be present at the point of VI for my conclusions to stand, see footnote 1.

(3)	a. gled-a-**telj** watch-TH-N 'spectator'	b. kov-a-**ač** forge-TH-N 'blacksmith'	c. pis-a-**ac** write-TH-N 'writer'
(4)	a. pozn-**av**-a-**telj** know-AV-TH-N 'expert'	b. prouč-**av**-a-**telj** study-AV-TH-N 'researcher'	c. reš-**av**-a-**telj** solve-AV-TH-N 'solver'
(5)	a. predsed-**av**-a-**ač** chair-AV-TH-N 'chair'	b. pred-**av**-a-**ač** lecture-AV-TH-N 'lecturer'	c. ugnjet-**av**-a-**ač** oppress-AV-TH-N 'oppressor'
(6)	a. prod-**av**-a-**ac** sell-AV-TH-N 'seller'	b. dar-o-d-**av**-a-**ac** gift-L-give-AV-TH-N 'giftgiver'	c. posl-o-d-**av**-a-**ac** job-L-give-AV-TH-N 'employer'

In (4)-(6), we see that *n* is separated from the root at least by the morpheme AV. Still, the choice of the *n*-allomorph (*-telj, -ač, -ac*) is sensitive to the identity of the root. In other words, each individual root is arbitrarily specified to occur with some *n*-allomorph; there are no phonological or semantic predictors for co-occurence with a particular *n*-allomorph in the synchronic grammar of BCS.

Now, whatever the exact structure of the agent nominals in (4)-(6), linearization is argued to occur before allomorph selection (Embick 2010), making available the linear representation ROOT-AV-(TH)-N. Since the choice of the *n*-allomorph depends on the identity of the root, across at least one linearly intervening morpheme with an overt exponent (AV), this BCS case violates the linear adjacency requirement on allomorphy proposed in Embick 2010.[3] Hence, linear adjacency cannot be a general locality constraint on allomorphy. In section 4, I discuss how this conclusion can be reconciled with data from other empirical domains where linear adjacency has been argued to play a role in allomorph selection.

2.1 A note on the identity of *-av* and locality domains

In the previous section, what mattered was only that AV is a morpheme. This was sufficient to argue against a linear adjacency requirement for allomorphy. In this section, I focus on the identity of the morpheme AV. If AV is necessarily part of the verbal extended projection, as on proposals that treat it as an "imperfectivizer" or an "eventizer", then the patterns in (4)-(6) are surprising because they seemingly violate the basic cyclicity constraint on allomorphy in (1). If the agent nominals in (4)-(6) necessarily include verbal structure (and hence the categorizer *v*), then (4)-(6) would involve a configuration like (7), in which the root is accessible to the second cyclic head (*n*). Here, I will show evidence that suggests *-av* is not necessarily part of the verbal extended projection and may instead be a root.

[3] The "with an overt exponent" part is important because Embick 2010 and related work allows the pruning of morphemes whose exponents would be null at VI.

(7) [*n* ... [*v* [$\sqrt{\text{ROOT}}$ ...]]]

First, it is worth noting that, in addition to *-av*, another element, *-iv*, appears in the derivation of imperfective/atelic verbs in BCS (8). This morpheme may also appear in the derivation of some (but not all) agent nominals based on the corresponding roots, cf. (9a-b).

(8) a. zatašk-a-ti
cover_up-TH-INF
'cover up'

b. zatašk-iv-a-ti
cover_up-IV-TH-INF
'be covering up'

c. uruč-i-ti
deliver-TH-INF
'deliver'

d. uruč-iv-a-ti
deliver-IV-TH-INF
'be delivering'

(9) a. zatašk-iv-a-ač
coverup-IV-TH-N
'cover up agent'

b. uruč-i-telj
serve-TH-N
'process server'

Note that, despite *-av*/*-iv* having a clearly imperfectivizing/atelicizing role in the verbal domain, there are no aspectual differences between agent nominals with and without this morpheme. For example, *zataškivač* 'cover up agent' in (9a) does not necessarily denote an agent of an atelic/imperfective event, nor does *uručitelj* 'process server' in (9b) necessarily denote an agent of a telic/perfective event. Instead, agent nominals based on different roots idiosyncratically include the morpheme *-av*/*-iv* or not, as shown in (9).[4] Usually, only one of the options–with or without the morpheme *-av*/*-iv*–is possible (e.g., **zatašk-a-ač*, **uruč-iv-a-telj*), but there are some exceptions, as seen in (10). Such examples are illuminating because they again show that there are no aspectual differences in the interpretation of agent nominals with and without *-av*/*-iv*. Hence, despite the seemingly obvious aspectual contributions of these morphemes in the verbal domain, there is reason to doubt that they are literal exponents of aspectual heads.

(10) a. ponuđ-a-ač
offer-TH-N
'supplier'

b. ponuđ-iv-a-ač
offer-IV-TH-N
'supplier'

Quaglia et al. (2022) follow the same intuition, arguing that *-av/-iv* are (exponents of) roots. They note that *-av/-iv* appear in the derivation of many nouns and adjectives for which a deverbal derivation is not likely because the corresponding verbs do not exist. For example, *-av* is found in the derivation of many adjectives (11a), and *-iv* is found both in (morphologically complex) nouns and adjectives (11b-c). For Quaglia et al., *-av/-iv* are bound roots, roots which may participate in the derivation of complex words, but which never appear categorized on their own. If *-av* and *-iv* are roots and not morphemes that

[4] It is worth noting that agent nominals are statistically much more often derived from what look like (primary or secondary) atelic/imperfective stems, but this is not without exception, as (9b) shows. I do not have an explanation for this statistical tendency.

necessarily appear in the verbal domain, then this explains why *n* in the agent nominals I discussed has access to the root: It is the first-merged categorizer.

(11) a. prlj-av-a
dirt-AV-F.SG
'dirty$_{(F)}$'

b. maz-iv-o
daub-IV-NEUT.SG
'grease'

c. jez-iv-a
shudder-IV-F.SG
'creepy$_{(F)}$'

Further evidence for this analysis comes from the accentual system of BCS. BCS is a pitch-accent language and accent is lexically determined, that is, idiosyncratically listed as part of some exponents of some morphemes, but not others (see Inkelas and Zec 1988). Only one accent is allowed to surface per prosodic word. Bešlin 2025 finds that exponents with an underlying accent can only realize it if they are spelled out in the first cyclic domain, where *cyclic domain* is understood as in (1). In case there are multiple underlying accents in the first cyclic domain, the structurally highest accent wins out. For example, the *n*-allomorph *-ac* is underlyingly accent-marked (*ác*), but only realizes that accent if it is the first-merged categorizer, regardless of whether the root is a bound root like *bor-* 'fight' in (12a), or not, like *škrt* 'stingy', which can be adjectivized via a null suffix (12b). The accent does not surface on the underlyingly accent-marked *ác* if *ác* attaches to an already categorized element, for example, in deadjectival nominals (13). In (13a) the accent surfaces on the root because the *a* (*-ov*) does not have an accent, while in (13b), the accent surfaces on the underlyingly accent-marked *a -ljív*; see Bešlin 2025: chapter 6 for detailed discussion. Crucially, the accent in (13) cannot surface on *n* because accent placement is determined in the first spell-out domain.[5]

(12) a. bor-ác
fight-N
'fighter'

b. škrt-ác
stingy-N
'scrooge'

(13) a. nobél-ov-ac
Nobel-A$_{\text{POSS}}$-N
'Nobel winner'

b. krad-ljív-ac
rob-A-N
'thief'

Unlike categorization, (root-)compounding does not have the effect of closing off the locality domain for the realization of the accent (14). Hence, when a categorizer like *-ác* is merged to a root-root compound, its accent surfaces on the resulting noun.[6] As with obvious cases of compounding, the accent surfaces on the categorizers *-ác*/*áč* in agent nominals which include the morphemes *-av*/*-iv* (15).[7]

[5] Non-categorizing morphemes in the extended projection of the first categorizer (DEG and NEG for *a*, DIM for *n*) can still affect the position of the accent, just like (1) predicts; see Bešlin 2025: chapter 6.

[6] [o] is the most common 'linking vowel' in root-root compounds in BCS (Babić 2002).

[7] The *n*-allomorph *-telj*, on the other hand, does not have an underlying accent, so the accent never surfaces on it, even in clear cases of root-derived nouns.

(14) a. led-o-lom-ác
ice-L-break-N
'ice-breaker'

b. min-o-bac-áč
mine-L-throw-N
'mine layer'

(15) a. prod-av-a-ác
sell-AV-TH-N
'seller'

b. zatašk-iv-a-áč
cover_up-IV-TH-N
'cover up agent'

This further suggests that *-iv* and *-av* are not specifically verbal morphology, and that there may not be any verbal structure in *-iv*/*-av* agent nominals (for a discussion of why agent/event entailments should not generally be taken as evidence for the presence of verbal structure, see Bešlin 2025: chapter 8).

3. Case study II: Negative comparative adjectives

Recall that, in addition to being subject to the locality principle in (1), allomorphy has been argued to be constrained by three ancillary mechanisms: (i) linear adjacency (Embick 2010, a.o.); (ii) structural adjacency (Adger, Béjar, and Harbour 2001, 2003, a.o.); and (iii) accessibility domains, a claim that roots are visible up to the first node above the first categorizer–Y, but not X in (1) (Moskal 2015). In the previous section, we saw that BCS agentive nouns show patterns of allomorphy which violate (i), the linear adjacency requirement. In this section, I discuss negative comparative adjectives in BCS and show that the root and the comparative morpheme are visible to each other for allomorphy purposes in the structural configuration in (16). This calls into question the relevance of both structural adjacency and accessibility domains as general locality constraints in the grammar.

(16) [CMPR [NEG [*a* [ROOT]]]]

Turning now to comparative adjectives in BCS, the form of the comparative (*-ij-, -j-, -š-,* or -∅-/root suppletion) is not fully predictable in the synchronic grammar of BCS (Stanojčić and Popović 1992). Consider, for example, adjectives derived from short (17) and long (18) monosyllabic roots.[8] The choice of the comparative allomorph is not phonologically predictable; we are dealing here with suppletive, morphologically-conditioned (i.e., root-determined) allomorphy.

(17) a. sit 'full' + -ij-(i) 'CMPR-(M.SG)' → sitiji 'fuller'
b. strog 'fast' + -j-(i) 'CMPR-(M.SG)' → stroži 'faster'
c. mek 'soft' + -š-(i) 'CMPR-(M.SG)' → mekši 'softer'
d. zao 'bad' + -∅-(i) 'CMPR-(M.SG)' → gori 'worse'

[8] The suffix *-j-(i)* triggers a regular palatalization (iotation) process, such that e.g., /gj/ → [ʒ].

(18) a. slaan 'salty' + -ij-(i) 'CMPR-(M.SG)' → slaniji 'saltier'
b. jaak 'strong' + -j-(i) 'CMPR-(M.SG)' → jači 'stronger'
c. leep 'pretty' + -š-(i) 'CMPR-(M.SG)' → lepši 'prettier'
d. maal- 'small' + -∅-(i) 'CMPR-(M.SG)' → manji 'smaller'

Negated adjectives can also be used in the comparative form, despite the relative infrequency of naturally occurring examples in the corpus, which is most likely due to pragmatic factors.[9] Negative comparative adjectives are nonetheless available with all four of the above-mentioned comparative allomorphs, see (19),[10] (20),[11] (21),[12] and (22).[13]

(19) Mladi sir koji sam jutros kupila je pun
young cheese which AUX this_morning bought is full
pogodak... odavno ni-je bio blaži, mlađi i
score a_while NEG-AUX was milder younger and
ne-slan-ij-i.
NEG-salty-CMPR-M
'The young cheese I bought this morning is a huge hit... It hasn't been milder, younger and unsaltier than this in a while.'

(20) Budući da se "ficklewise" ne može lako prevesti
given that SE ficklewise NEG can easily translate
zbog svoje **ne-čest-j-e** upotrebe, moglo bi
because its NEG-frequent-CMPR-F.GEN use could AUX
se prevesti kao "promjenljivo" ili "neodlučno" u kontekstu
SE translate as changeable or indecisive in context
mudrosti ili razmišljanja.
wisdom or thinking
'Given that *ficklewise* cannot be easily translated due to its more infrequent use, it could be translated as *changeable* or *indecisive* in the context of wisdom or thinking.'

[9] For example, in order to use *ne-bolj-e* 'more ungood', as in (22), the speaker must want to make a distinction between *nebolje* 'more ungood' and the more common *gore* 'worse', despite their truth-conditional equivalence. In (22), the naturalness of the negative comparative is arguably increased by the use of the comparative form in the preceding clause.
[10] Available: `https://tinyurl.com/mvvyd5um`
[11] Available: `https://tinyurl.com/yn877t9x`
[12] Available: `https://tinyurl.com/yc2a57h3`
[13] Available: `https://tinyurl.com/2kdczspp`

(21) Hvala svima koji ste mi učinili dan lep-š-im
thanks everyone who AUX me made day pretty-CMPR-INSTR.M
čestitkama i dolaskom na rođendanski stendap, a
well-wishes and coming to birthday stand-up and
hvala i onima koji su ne-javljanjem učinili da
thanks also those who AUX NEG-calling made that
dan ne bude **ne-lep-š-i**.
day NEG be NEG-pretty-CMPR-M
'Thanks to everyone who made my day more beautiful with their well-wishes and by coming to my birthday stand-up, and thanks also to those who, by not calling, made it so my day was not more unbeautiful.'

(22) Ali bit će bolje i strah te, šta ako ti to
but be FUT better and fear you what if you that
bolje dođe, ali ti ne bude dobro jer se
better comes but you NEG be good because SE
navikneš na ovo **ne-bolje**. Ja ne volim ovo **ne-bolje**...
accustom to this NEG-better I NEG love this NEG-better
'But it will get better and you're afraid, what if the better comes and you are not good because you are used to this worse (situation). I don't like this worse (situation)...'

In all of the examples of negative comparatives presented here, the comparative scopes over the negation (CMPR>NEG); had the negation scoped over the comparative (NEG>CMPR), the interpretation would have been weaker. Let's unpack this using a constructed example (23). The interpretation of the negative comparative is straghtforwardly CMPR>NEG: The second cheese is less salty (i.e., more unsalty) than the first. While the stronger reading obtained in CMPR>NEG entails the weaker NEG>CMPR, the reverse entailment relation does not hold. In other words, while being less salty entails not being more salty, not being more salty does not entail being less salty. Had NEG outscoped CMPR, the reading we obtain in a sentence like (23) would have been unavailable. Hence, the structure of such negative comparative adjectives in BCS is as in (24), with the comparative morpheme outscoping negation.

(23) **Context**: At the market, I want to buy cheese that's not too salty. The seller says:
Ovaj je dosta ne-slan. A ovaj je još **ne-slan-ij-i**.
this is pretty NEG-salty and this is even NEG-salty-CMPR-M
'This one is pretty unsalty. And this one is even more unsalty (than the first).'

(24) [φ -i [CMPR -ij- [NEG ne- [a -an [$\sqrt{\text{SL}}$]]]]]

Recall that we have shown the form of the comparative allomorph to be dependent on the identity of the root it combines with. Given that conclusion, CMPR must be able to see the root (qua morpheme) despite the intervention of NEG and *a*. This BCS case therefore violates the structural adjacency condition on allomorphy (contra Adger, Béjar, and Harbour 2001, 2003, a.o.). Since at least NEG is always overt (and *a* only sometimes), this conclusion holds regardless of whether only overt morphemes are counted for purposes of the adjacency requirement. Additionally, access to the root is not restricted to the first node above the categorizer, contra Moskal 2015. As seen in (24), CMPR is the second node above the categorizer and it is still able to see the root for purposes of allomorph selection.[14]

3.1 On bracketing paradoxes

In this section, I address the worry that negative comparative adjectives may not be the best testing ground for locality theories due to their controversial structure, discussed under the rubric of 'bracketing paradoxes'. I show that bracketing paradoxes do not exist for BCS negative comparative adjectives. In English, the observation has been that for some negative comparative adjectives the correct constituency for form purposes is in (25a), while the correct meaning is derived with the structure in (25b). The constituency in (25a) seems needed for form purposes, because only mono- and disyllabic adjectives in English may combine with the comparative allomorph *-er*. Hence, the adjective *happy* seems to combine with the comparative morpheme before the addition of the negative prefix. Had the trisyllabic *unhappy* combined with the comparative morpheme, we would expect the comparative allomorph *more* to surface obligatorily. On the other hand, it is clear from the meaning of *unhappier* that the comparative should scope over the negation, as in (25b).

(25) a. [un- [happi-er]] b. [[un-happy] -er]

Notice that this problem is quite specific to English, given the apparent syllable-counting property of the comparative morpheme. To address this issue, Newell (2008) argues that the structural relations between the morphemes in *unhappier* are (ultimately) as in (25b), but where *un-* is an adjunct which attaches acyclically to the bottom of the tree after spell-out of the adjective *happy* and the comparative morpheme to the form interface. In this way, no bracketing paradox arises.

This account may be needed for English negative comparatives but it is both unnecessary and demonstrably false in BCS. As I discussed in section 2.1, BCS accent is an idiosyncratic property of individual exponents. The accent gets realized on the structurally highest underlyingly accent-marked element in the first spell-out domain, where the first spell-out domain is understood as in (1). Non-categorizing morphemes in the extended projection of the first categorizer can still affect the position of the accent, just like (1) predicts. For example, both *né-* 'NEG' and *-íj* 'CMPR' have an underlying accent, so pitch prominence surfaces on them in adjectives that contain them, (26) (cf. *slán* 'salty').

(26) a. né-sl-an
NEG-salt-A
'unsalty'

b. sl-an-íj-i
salt-A-CMPR-M
'saltier'

[14] For Moskal 2015, the overtness of the various morphemes is not supposed to play a role; categorizers are usually phonologically empty in her examples.

Note that the accent shifts to the position of the highest underlyingly accent-marked exponent (in the first spell-out domain) regardless of that exponent's linear position as a prefix or a suffix, supporting the idea that the position of the accent is determined in structural terms. Crucially for the discussion of bracketing paradoxes, accent, which is calculated within the first spell-out domain in BCS, surfaces on adjectival NEG in negative adjectives (26a), strongly suggesting NEG is merged in the first cycle, and not acyclically as Newell (2008) suggests for English.

We can additionally use the position of the accent to further corroborate the proposed structure of the BCS negative comparative adjective, given in (24) and repeated here in (27). Recall the claim that pitch prominence surfaces on the structurally highest accent-marked exponent in the first spell-out domain. In negative comparatives, we then predict the accent to surface on the comparative when both it and negation co-occur on an adjective. The prediction is borne out (28).

(27) [φ -i [CMPR -ij- [NEG ne- [*a* -an [$\sqrt{\text{SL}}$]]]]]

(28) ne-sl-an-íj-i
NEG-salt-A-CMPR-M
'unsaltier'

Given all this, we can conclude that here is no bracketing paradox in BCS negative comparative adjectives. Instead, both the form and the meaning interface are clearly fed the structure in (24)/(27).

4. Conclusions and implications for the ordering of VI and linearization

In this paper, I have upheld the claim that cyclic spellout plays a role in determining which portions of syntactic structure are accessible at the form interface at any given time. The main goal was to show that the ancillary locality mechanisms argued to play a role at VI should not be regarded as universal. I showed that BCS exhibits patterns of allomorphy which violate both linear and structural adjacency requirements, as well as undermine the broad utility of accessibility domains. This is not the first work to report data which is problematic for each of these mechanisms (see, for example, Kastner and Moskal 2018, a special issue of *Snippets* dedicated entirely to non-local allomorphy). This paper does, however, show an example of a language where *all* and not just some of the proposed mechanims can be shown not to be operative in at least some contexts.

If we can do away with these ancillary locality constraints, we can simplify the grammar in one of two ways. If the data reported thus far in support of these constraints can be reanalyzed, then the mechanisms that enforce them may not be part of the grammar at all. There is value in attempting such alternative analyses in future work. However, even if it turns out that a notion like adjacency is indisputably referenced by some VI rules, this does not mean that adjacency is a general constraint on VI contexts. If certain VI rules require adjacency, this may be learnable given positive input. Still, adjacency requirements do not seem to be a universal property of VI contexts, given the data I have discussed here.

Illustrating schematically with (29), note that the BCS data instantiate a pattern where the morpheme Z conditions allomorphy on X over a closer intervening morpheme Y, but Y itself does not condition allomorphy on X. There is, however, a robust generalization in the literature that Z may not condition allomorphy on X over an intervener Y if Y itself conditions allomorphy on X (*ABA, Bobaljik 2012).[15] We may therefore still need a general locality principle on VI (in addition to cyclic spellout), one based on intervention. If VI applies from the root outwards (Bobaljik 2000, Kalin and Weisser to appear), then such a principle would ensure that, for a root X in (29), the closer morpheme Y always wins out as a contextual trigger over the further away Z if both can in principle trigger allomorphy, in turn accounting for *ABA.

(29) X Y Z

Still, the question remains whether the 'closer' morpheme here is calculated in structural or linear terms. If linearization applies before VI (Embick 2010, Wood 2015, Kalin 2022, a.o.), this still in principle leaves both options open. As argued in Embick 2010 and Wood 2015, linearization is a two-step process. The grammar first produces a bracketed linear structure which preserves the hierarchical ordering (30a), and only after this is the concatenation operation executed to output a purely linear string (30b).

(30) a. (X * (Y * Z)) b. $X \frown Y \frown Z$

If VI occurs after the first step in (30a), then VI should in principle be able to reference both linear and hierarchical information. There is indeed evidence that both linear and structural information can be referenced at VI (e.g., Bobaljik 2000, Embick 2010, Wood 2015, Kalin 2022, Bešlin 2025). This suggests that VI occurs after linearization but before concatenation (i.e., before structural information is lost).

An understanding of linearization as a two-step process, as in (30), may also give us a handle on why certain VI rules are able to reference linear adjacency without necessarily making this a universal feature of the system. Specifically, if linearization makes available the notion of linear adjacency, then that information may become available at the point of bracketed linear representations, as in (30a). If bracketed linear representations are furthermore obtained before VI as seems necessary to accommodate the data in the literature, then individual VI rules *may* (but need not) refer to linear adjacency.

[15] In Bobaljik's terms, the BCS cases I discussed instantiate an AAB pattern. According to Bobaljik, there is no general ban on AAB: It is attested in some domains and unattested in others.

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